

Malware Analysis Lab

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Task 1 – Static Analysis (PEStudio 9.51)

Instructions: Open **hello.exe** in PEStudio and record the following:

Architecture: PE32+ for 64-bit(executable, 64-bit, GUI)

Entry Point (EP): 0x00021AA0(section: UPX1)

Number of Imports: 6 imports.

Key Imports (list 2–3)

- VirtualProtect (*)
- LoadLibraryA & GetProcAddress
- MessageBoxA

Sections (VA, Virtual Size, Raw Size / Offset, Permissions)

- 0x00001000 Virtual Address
- 0x00019000 (102,400 bytes) Virtual Size
- Write, Execute and Self-Modifying.

Task 2 – Dynamic/Hybrid Analysis (VirusTotal)

Instructions: Upload the file or SHA256 hash to VirusTotal and record observations.

Detection Ratio: 42/72

Example AV Names

- ESET-NOD32
- Microsoft
- CrowdStrikefalcon

MITRE Tactics and Techniques

- Tactic: Defense Evasion (TA0005)

- Tactic: Execution (TA0002)

What tool was used to pack this software for defensive evasion? UPX

Task 3 – Reflection

Instructions: Summarize your findings in 3–5 sentences, comparing static and dynamic analysis.

Reflection: Based on the architectural indicators, static analysis initially identified this file as a 65-bit(PE32+) executable packed with UPX, which effectively obscured its true functionality by hiding the majority of its imports and strings. While static tools flagged suspicious elements like VirtualProtect and a high 42/72 VirusTotal score, they could only see the “outer shell” or unpacking stub rather than the malicious payload. In contrast, dynamic analysis would allow the malware to decompress into the UPX0 section, revealing its actual behavior. Hidden API calls and communication patterns in real time. Comparing the two reveals a defense evasion strategy where the file's disk-based blueprint is intentionally minimal to force an analyst into a more complex, time consuming dynamic investigation.